

Lecture 18: UI Duration and Job Quality – Live In-Class Empirical Exercise

Compiled: April 22, 2026

Exercise Overview

This handout is built from `Lecture18_uidiwc_liveanalysisinclass_dy.do` and includes only the outputs that script actually generates.

- Starting file: `Lecture18_uidiwc_liveanalysisinclass_dy.do`
- Main input: `Data_public.dta`
- Identification idea: Austrian UI duration jumps at age 40, so the figures plot outcomes around that threshold
- This shortened script exports Figure 1 and Figure 3, then writes the Stata log shown at the end of this PDF

Stata Code

```
*
* *****
*
* LECTURE 18: UI DURATION AND JOB QUALITY
* LIVE IN-CLASS REPLICATION OF FIGURES 1, 2, AND 3
* Nekoei and Weber (2017, AER)
*
* *****
*
* Design in one sentence:
* Austria extended UI benefit duration discontinuously at age 40, so the
* paper
* studies whether outcomes jump at that age threshold.
*
* Policy in plain English:
* Workers below age 40 are eligible for 30 weeks of UI, while workers at
* age
* 40 and above are eligible for 39 weeks. That nine-week jump creates the
* left-
* versus-right comparison in the figures. The plots check whether outcomes
* and
```

```

* predetermined characteristics stay smooth apart from the policy break.
*
* Main figures kept up front:
* Figure 1: Sorting / density check around age 40
* Figure 2: Pre-reform placebo check
* Figure 3: Main outcome plot
*
* Extra plots are moved to the end of the file.
*
*****

version 14
clear all
set more off

*
*****

* 0) Paths
*
*****

if "`c(username)'" == "muhammadbashir" {
    global INCLASS_DIR "/Users/muhammadbashir/Library/CloudStorage/
        Dropbox/ugradteaching/econ131spring26/LectureEmpiricalExercises
        /Lecture18_uidiwc_ch14/inclass"
}
else if "`c(username)'" == "Danny" {
    global INCLASS_DIR "C:/Users/Danny/Dropbox/ugradteaching/
        econ131spring26/LectureEmpiricalExercises/Lecture18_uidiwc_ch14
        /inclass"
}
cd "$INCLASS_DIR"

capture log close _all
log using "Lecture18_uidiwc_liveanalysisinclass_output.log", text replace
    name(uidiwc18)

*
*****

* 1) Definitions
*
*****

* Shared constants are defined as globals. Plot-specific variables are
    generated
* in the sections below right next to the relevant figure.
global age_cutoff 40 // UI threshold age where benefit duration changes
global age_bins_per_year 3 // number of age bins per year in the scatter
    plots
global age_bin_midpoint 1/6 // center point of each age bin

```

```

global ui_max_duration 2*365 // two years in days
global long_job_window 39*7 // 39 weeks in days
global half_wage_ln ln(.5) // log of one-half

* Variables used in the plots:
* agec = age binned into thirds of a year, centered within each bin
* id0 = first observation in each age bin, used for scatter overlays
* lwage0 = ln(monthly_wage_0)
* lwagen0 = ln(monthly_wage_n0)
* ay1 = non-employment duration, restricted to spells shorter than 2 years
* ay3 = indicator for finding a job within 39 weeks
* ay4 = wage change, restricted to spells shorter than 2 years
* ay6 = indicator that the new wage exceeds half of the old wage
* total_n = total sample size, used to normalize the age-density plot
* bin_n = within-bin count normalized by total sample size
* mean_lwage0 = mean previous log wage by age bin
* y = temporary plotting variable
* m = temporary bin mean used in the placebo and outcome plots

*
*****

* 2) VALIDATE THE DESIGN: ARE COUNTS SMOOTH ACROSS THE TREATMENT
  DISCONTINUITY?
*
*****

*LOAD THE AUSTRIAN DATA
use "../Nekoei_Weber_2017_UI_job_quality/20150528_data/Data/Data_public.
  dta", clear

*DO BASIC VARIABLE PREPARATION
gen agec = (int(age*$age_bins_per_year)/$age_bins_per_year) +
  $age_bin_midpoint // age binned into thirds of a year
bysort agec: gen id0 = (_n == 1) // mark the first observation in each age
  bin
gen total_n = _N // total sample size for normalization
bysort agec: gen bin_n = _N // count observations within each age bin
replace bin_n = bin_n / total_n // convert bin counts to sample shares

*GRAPH THE AGE DISTRIBUTION OF LAID OFF WORKERS IN AUSTRIA
twoway ///
  (scatter bin_n agec if id0, mcolor(cranberry) msize(small)) ///
  (qfit bin_n agec if id0 & age > $age_cutoff, lcolor(navy) lwidth(
    medthick)) ///
  (qfit bin_n agec if id0 & age < $age_cutoff, lcolor(navy) lwidth(
    medthick)), ///
  xline($age_cutoff, lpattern(dash) lcolor(gs10)) ///
  xtitle("Age at layoff") ytitle("Density") legend(off) ///
  graphregion(fcolor(white)) ///
  subtitle("Main figure 1. Distribution of age", size(small)) ///
  name(fig1_main, replace) // save the main age-density figure
graph export "figure01_uidiwc.pdf", replace // export main plot 1

```

```

*
*****

* 3) USE THE DESIGN: DOES THE PROBABILITY OF FINDING A JOB FALL AT THE
  TREATMENT DISCONTINUITY?
*
*****

*LOAD THE AUSTRIAN DATA
use "../Nekoei_Weber_2017_UI_job_quality/20150528_data/Data/Data_public.
dta", clear

*DO BASIC VARIABLE PREPARATION
gen agec = (int(age*$age_bins_per_year)/$age_bins_per_year) +
  $age_bin_midpoint // age binned into thirds of a year
bysort agec: gen id0 = (_n == 1) // mark the first observation in each age
  bin
gen ay3 = ned <= $long_job_window // indicator for finding a job within 39
  weeks
gen y = ay3 // temporary plotting variable for the job-finding panel
egen m = mean(y), by(agec) // bin mean used in the scatter plot

*GRAPH THE PROBABILITY OF FINDING A JOB WITHIN 39 WEEKS VS THE AGE AT
  LAYOFF IN AUSTRIA
twoway ///
  (scatter m agec if id0, mcolor(cranberry) msize(small)) ///
  (qfit y age if age >= $age_cutoff, lcolor(navy) lwidth(medthick))
  ///
  (qfit y age if age < $age_cutoff, lcolor(navy) lwidth(medthick)),
  ///
  xline($age_cutoff, lpattern(dash) lcolor(gs10)) ///
  xtitle("Age_at_layoff") ytitle("Prob. of finding job within 39
  weeks") ///
  legend(off) graphregion(fcolor(white)) ///
  subtitle("Panel B. Main outcome: Job finding within 39 weeks",
    size(small)) ///
  name(fig3b_extra, replace) // save the extra job-finding plot
graph export "figure03_uidiwc.pdf", replace // export main plot 3

```

Figure 1

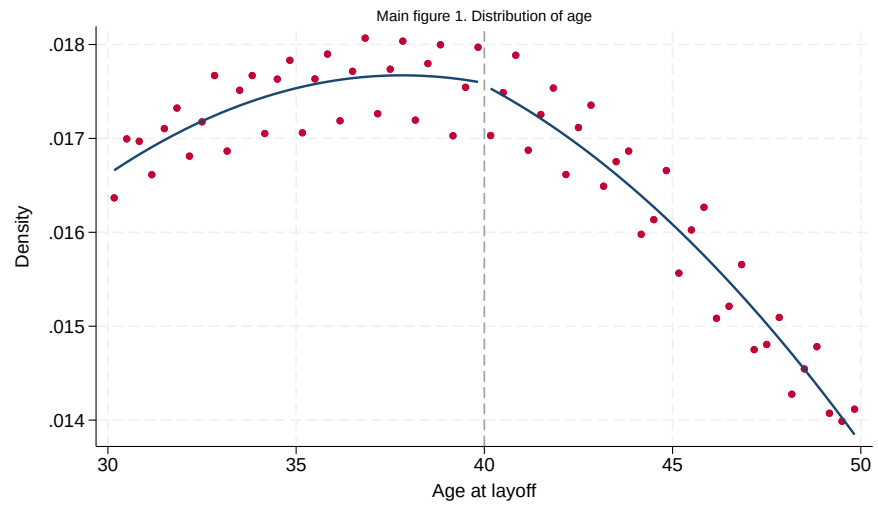
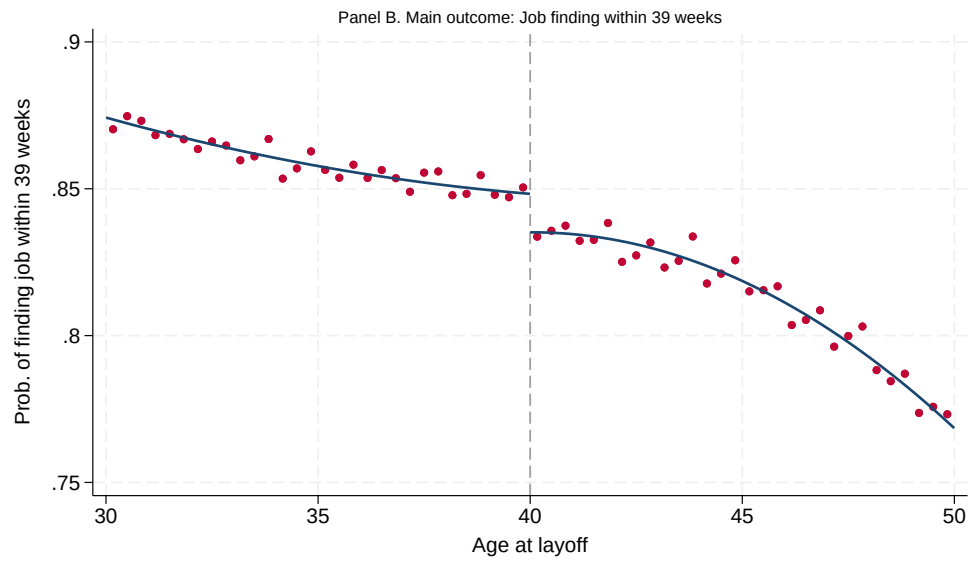


Figure 3



```

-----
name: uidiwc18
log: /Users/muhammadbashir/Library/CloudStorage/Dropbox/ugradteaching/e
> con131spring26/LectureEmpiricalExercises/Lecture18_uidiwc_ch14/inclass/Lecture
> e18_uidiwc_liveanalysisinclass_output.log
log type: text
opened on: 22 Apr 2026, 16:33:49
.
.
.
. *****
> ***
. * 1) Definitions
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. global ui_max_duration 2*365 // two years in days

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. * agec = age binned into thirds of a year, centered within each bin
. * id0 = first observation in each age bin, used for scatter overlays
. * lwage0 = ln(monthly_wage_0)
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. * ay1 = non-employment duration, restricted to spells shorter than 2 years
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> clear

.
. *DO BASIC VARIABLE PREPARATION
. gen agec = (int(age*$age_bins_per_year)/$age_bins_per_year) + $age_bin_midpoi
> nt // age binned into thirds of a year

. bysort agec: gen id0 = (_n == 1) // mark the first observation in each age bi
> n

```

```

. gen total_n = _N // total sample size for normalization

. bysort agec: gen bin_n = _N // count observations within each age bin

. replace bin_n = bin_n / total_n // convert bin counts to sample shares
(1,738,787 real changes made)

.
. *GRAPH THE AGE DISTRIBUTION OF LAID OFF WORKERS IN AUSTRIA
. twoway ///
>     (scatter bin_n agec if id0, mcolor(cranberry) msize(small)) ///
>     (qfit bin_n agec if id0 & age > $age_cutoff, lcolor(navy) lwidth(medt
> hick)) ///
>     (qfit bin_n agec if id0 & age < $age_cutoff, lcolor(navy) lwidth(medt
> hick)), ///
>     xline($age_cutoff, lpattern(dash) lcolor(gs10)) ///
>     xtitle("Age at layoff") ytitle("Density") legend(off) ///
>     graphregion(fcolor(white)) ///
>     subtitle("Main figure 1. Distribution of age", size(small)) ///
>     name(fig1_main, replace) // save the main age-density figure

. graph export "figure01_uidiwc.pdf", replace // export main plot 1
file figure01_uidiwc.pdf saved as PDF format

.
.
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>     (scatter m agec if id0, mcolor(cranberry) msize(small)) ///
>     (qfit y age if age >= $age_cutoff, lcolor(navy) lwidth(medthick)) ///
>     (qfit y age if age < $age_cutoff, lcolor(navy) lwidth(medthick)), ///
>     xline($age_cutoff, lpattern(dash) lcolor(gs10)) ///
>     xtitle("Age at layoff") ytitle("Prob. of finding job within 39 weeks"
> ) ///
>     legend(off) graphregion(fcolor(white)) ///
>     subtitle("Panel B. Main outcome: Job finding within 39 weeks", size(s
> mall)) ///
>     name(fig3b_extra, replace) // save the extra job-finding plot

. graph export "figure03_uidiwc.pdf", replace // export main plot 3

```



```
file figure03_uidiwc.pdf saved as PDF format
```

[illegible]

```
end of do-file
```